DECIMAL, BINARY, AND HEXADECIMAL

CSE 101 Section 03
Introduction to Algorithms and Computing
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Decimal Numbering System

- Ten symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- Represent larger numbers as a sequence of digits
  - Each digit is one of the available symbols

- <u>Example</u>: 7061 in decimal (base 10)

$$7061_{10} = (7 \times 10^3) + (0 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$$

Octal Numbering System

- Eight symbols: 0, 1, 2, 3, 4, 5, 6, 7
  - Notice that we no longer use 8 or 9
- Base Comparison:
  - Base 10: 0, 1, 2, 3, 4, 5, 6, 7, 8 , 9, 10, 11, 12…
  - Base 8: 0, 1, 2, 3, 4, 5, 6, 7, 10 , 11, 12, 13, 14…

- <u>Example</u>: What is 15_8 in base 10?
  - $15_8 = (1 \times 8^1) + (5 \times 8^0) = 13_{10}$
- <u>Example</u>: What is 7061_8 in base 10?
  - $7061_8 = (7 \times 8^3) + (0 \times 8^2) + (6 \times 8^1) + (1 \times 8^0) = 3633_{10}$

Question

What is 34_8 in base 10?

(A) 32_{10}

(B) 34_{10}

(C) 7_{10}

(D) 28_{10}

(E) 35_{10}

Binary Numbering System

- Binary is base 2
  - Symbols: 0, 1
- Convention: $2_{10} = 10_2 = 0b10$

Base 10	Base 8	Base 2
0	0	0
1	1	1
2	2	10
3	3	11
4	4	100
5	5	101
6	6	110
7	7	111
8	10	1000
9	11	1001

- Example: What is 0b110 in base 10?
  - $0b110 = 110_2 = (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) = 6_{10}$

Hexadecimal Numbering System

- Hexadecimal is base 16 (>10)
  - Symbols? 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Convention: $14_{10} = E_{16} = 0xE$

- Example: What is 0xA5 in base 10?
  - $0xA5 = A5_{16} = (10 \times 16^1) + (5 \times 16^0) = 165_{10}$

  When converting hexadecimal to decimal, you treat the letters as their corresponding decimal values

Base 10	Base16
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

Question

Which of the following orderings is correct?

(A) 0xC < 0b1010 < 11

(B) 0xC < 11 < 0b1010

(C) 11 < 0b1010 < 0xC

(D) 0b1010 < 11 < 0xC

(E) 0b1010 < 0xC < 11

BASE CONVERSION

Converting to Base 10

Can convert from any base *to* base 10

- $110_2 = (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) = 6_{10}$
- $0xA5 = A5_{16} = (10 \times 16^1) + (5 \times 16^0) = 165_{10}$

We learned to think in base 10, so this is fairly natural for us.

<u>Challenge</u>: Convert into other bases (e.g. 2, 16)

Challenge Question

Convert 13_{10} to binary

Hints:

- $2^3 = 8$
- $2^2 = 4$
- $2^1 = 2$
- $2^0 = 1$

Converting from Decimal to Binary

Given a decimal number N:

- List increasing powers of 2 from *right to left* until ≥ N
- From *left to right*, ask is that (power of 2) ≤ N?
  - If YES, put a 1 below and subtract that power from N
  - If NO, put a 0 below and keep going

Example for 13

$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0				

Converting from Decimal to Binary

Given a decimal number N:

- List increasing powers of 2 from *right to left* until ≥ N
- From *left to right*, ask is that (power of 2) ≤ N?
  - If YES, put a 1 below and subtract that power from N
  - If NO, put a 0 below and keep going

Example for ~~13~~
5

$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1			

Converting from Decimal to Binary

Given a decimal number N:

- List increasing powers of 2 from *right to left* until ≥ N
- From *left to right*, ask is that (power of 2) ≤ N?
  - If YES, put a 1 below and subtract that power from N
  - If NO, put a 0 below and keep going

Example for ~~13~~

~~5~~

1

$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1	1		

Converting from Decimal to Binary

Given a decimal number N:

- List increasing powers of 2 from *right to left* until ≥ N
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  - If YES, put a 1 below and subtract that power from N
  - If NO, put a 0 below and keep going

Example for ~~13~~

~~5~~

1

$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1	1	0	

Converting from Decimal to Binary

Given a decimal number N:

- List increasing powers of 2 from *right to left* until ≥ N
- From *left to right*, ask is that (power of 2) ≤ N?
  - If YES, put a 1 below and subtract that power from N
  - If NO, put a 0 below and keep going

Example for ~~13~~

~~5~~

~~1~~

0

$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1	1	0	1

Converting from Decimal to Base B

Given a decimal number N:

- List increasing powers of B from right to left until ≥ N
- From left to right, ask is that (power of B) ≤ N?
  - If <u>YES</u>, put *how many* of that power go into N and subtract from N
  - If <u>NO</u>, put a 0 and keep going

Example for ~~**165**~~ into hexadecimal (base 16):

~~**5**~~

0

$16^2 = 256$	$16^1 = 16$	$16^0 = 1$
0	A (10)	5

Converting Binary ↔ Hexadecimal

Hex → Binary

- Substitute hex digits, then drop leading zeros
- Example: 0x2D in binary
  - 0x2 is 0b0010, 0xD is 0b1101
  - Drop two leading zeros, answer is 0b101101

Base 10	Base 16	Base 2
0	0	0
1	1	1
2	2	10
3	3	11
4	4	100
5	5	101
6	6	110
7	7	111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Converting Binary ↔ Hexadecimal

Binary → Hex

- Pad with leading zeros until multiple of 4, then substitute groups of 4
- Example: 0b101101
  - Pad to 0b00101101
  - Substitute to get 0x2D

Base 10	Base 16	Base 2
0	0	0
1	1	1
2	2	10
3	3	11
4	4	100
5	5	101
6	6	110
7	7	111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Binary → Hex Practice

Convert 0b1001101101010101

- How many digits? 15
- Pad: 0b 0100 1101 1010 1101
- Substitute: 0x4DAD

Base 10	Base 16	Base 2
0	0	0
1	1	1
2	2	10
3	3	11
4	4	100
5	5	101
6	6	110
7	7	111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111